

PROJECT PROPOSAL

Smart Pizza Shop Management System

Project Type	Commercial-Oriented Python Software Application
Programming Language	Python
Database	SQLite
Programming Approach	Object-Oriented Programming (OOP)
User Interface	Graphical User Interface (GUI)

1. Introduction

The **Smart Pizza Shop Management System** is a commercial-oriented software application designed to help small and medium-sized pizza shops manage their daily operations more efficiently.

The main purpose of this project is to develop a practical system that goes beyond simple order registration. The application will provide a graphical user interface (GUI) for managing products, customers, orders, ingredients, inventory, sales, and business information.

The system will also include a smart inventory and recommendation feature that can analyze product recipes, ingredient consumption, and current stock levels to help the shop owner make better inventory decisions.

2. Problem Statement

Managing a pizza shop manually can cause several problems, such as incorrect order records, difficulty tracking inventory, wasted ingredients, and lack of accurate information about sales and popular products.

For example, when several pizzas are sold, the shop owner may not know exactly how much cheese, meat, mushrooms, or other ingredients have been consumed. As a result, ingredients may run out unexpectedly or be purchased in unnecessary quantities.

This project aims to solve these problems by connecting the ordering system with product recipes and inventory management.

3. Project Objectives

- To develop a practical software system for managing a pizza shop.
- To provide an easy-to-use graphical user interface.
- To manage pizzas, prices, sizes, and ingredients.
- To manage customers and their order history.
- To manage orders and calculate their total prices.
- To automatically update ingredient inventory after orders.

- To identify ingredients with low stock.
- To provide useful sales and inventory reports.
- To provide recommendations for restocking ingredients.
- To create a system that has potential for real-world commercial use.

4. Main Features

4.1 Product Management

- Add new pizzas.
- Edit and delete existing pizzas.
- Define different pizza sizes and prices.
- Define the ingredients required for each pizza.
- View available products.

Each pizza can have its own recipe. For example: **Special Pizza** — 200 g Mozzarella Cheese, 150 g Meat, 100 g Mushrooms, and 100 g Sausage.

4.2 Inventory Management

- Store ingredients and their quantities.
- Record incoming inventory.
- Reduce inventory when an order is completed.
- Display current stock.
- Identify low-stock ingredients.
- Show ingredients that need to be restocked.

This feature connects the sales process directly to inventory management.

4.3 Order Management

- Create a new order.
- Select products and quantities.
- Select the customer.
- Calculate the total price automatically.
- View order details.
- Change the order status.

Possible order statuses: Registered, Preparing, Ready, Completed, and Cancelled.

4.4 Customer Management

- Add customers.
- Edit customer information.
- Search for customers.
- View previous orders.

- Display customer purchase history.

5. Innovative Feature

The main innovative aspect of the project will be a **Smart Inventory and Restocking Recommendation System**.

Unlike a basic restaurant management system that only stores inventory information, this system will use order data and pizza recipes to estimate ingredient consumption.

For example, if customers order several pizzas, the system can calculate how much cheese, meat, mushrooms, and other ingredients have been used.

If an ingredient reaches a defined minimum stock level, the system will display a warning.

Example: Low Stock Warning — Mozzarella Cheese: 2.5 kg remaining — Recommended action: Restock.

The system can also analyze recent sales information and provide a simple restocking recommendation based on previous consumption.

This feature is intended to help reduce ingredient shortages and unnecessary inventory.

6. Business Dashboard

The application will include a graphical dashboard where the shop owner can quickly see important business information.

- Today's total sales.
- Number of orders.
- Best-selling pizza.
- Current low-stock ingredients.
- Recent orders.
- Inventory status.

Example: Today's Sales: €850 Orders: 37 Best Seller: Special Pizza Low Stock Items: 3

7. Graphical User Interface

The system will use a **Graphical User Interface (GUI)** instead of a console-only interface.

The main interface will contain separate sections for Dashboard, Orders, Products, Ingredients, Inventory, Customers, and Reports.

The interface will be designed to be simple and easy to use for a shop employee or owner who may not have technical knowledge.

8. Technologies

- **Programming Language:** Python
- **GUI:** Python GUI framework
- **Database:** SQLite
- **Programming Approach:** Object-Oriented Programming (OOP)

The exact GUI framework will be selected according to the project requirements and the technologies appropriate for the course.

9. Proposed Classes

- Pizza
- Ingredient
- Recipe
- Customer
- Order
- OrderItem
- Inventory
- Payment
- Report

The final class structure will be refined during the development process.

10. Database

SQLite will be used to store the application's persistent data.

The database may contain tables such as:

- pizzas
- ingredients
- recipes
- customers
- orders
- order_items
- inventory
- payments

The database structure will be designed to maintain relationships between products, recipes, orders, and ingredients.

11. Commercial Potential

The project is designed with real-world commercial use in mind.

A small pizza shop could potentially use such a system to:

- Manage daily orders.
- Monitor ingredients.
- Reduce inventory waste.
- Track sales.

- Identify popular products.
- Make better purchasing decisions.

The application can also be expanded in the future with additional features such as online ordering, delivery management, employee management, customer loyalty programs, and more advanced sales analysis.

12. Future Development

- Online customer ordering.
- Delivery management.
- Customer loyalty and discount systems.
- Employee management.
- Advanced sales analytics.
- Automatic purchase planning.
- Integration with online payment systems.
- Web or mobile versions of the application.

These features are not required for the initial version but can be considered for future development.

13. Expected Outcome

The expected result is a functional, user-friendly, and commercially oriented pizza shop management application.

The final system will combine: **Order Management + Product Management + Customer Management + Inventory Management + Sales Reports + Smart Restocking Recommendations.**

The project will demonstrate the practical application of Python programming, Object-Oriented Programming, database management, GUI development, and basic business-oriented decision support.

14. Conclusion

The **Smart Pizza Shop Management System** is intended to be more than a simple CRUD or restaurant ordering project. Its main goal is to create a practical software solution that can be useful in a real pizza shop.

By connecting orders with recipes and inventory, and by providing smart stock warnings and restocking recommendations, the system will offer additional value to the business owner.

The project will initially focus on the core features required for a small pizza shop while maintaining a structure that allows the application to be expanded into a more complete commercial product in the future.